

Manuscript title: *Linear probing enables near-perfect identification of bird species in acoustic recordings*

The author proposes a two-stage ("hybrid") approach to detecting Eurasian Woodcock (*Scolopax rusticola*) calls: BirdNET embeddings serve as fixed feature representations, and three secondary classifiers (SVM, Random Forest, XGBoost) are trained on top. Two strongly labelled datasets are used, one newly annotated from Xeno-canto (97 recordings; 501 positive and 2505 negative 3-second segments), and one built from Holderried (2024). BirdNET baseline average precision (AP) is reported as 84.3% on Dataset 1 and 89.3% on Dataset 2, rising to 99.8-100% once the secondary classifiers are applied, leading to the conclusion that the method achieves "near-perfect" species identification.

The topic is timely, the two-stage embedding-plus-classifier approach is well motivated and consistent with recent literature, and the manuscript is clearly structured. The publicly released strongly labelled annotations are a genuine contribution to woodcock monitoring, and the baseline BirdNET evaluation is worthwhile in its own right.

However, several choices in dataset construction and evaluation appear likely to inflate the reported performance, and the near-100% figures may reflect artefacts of the protocol rather than genuine detection capability. Three concerns are principal, and each could plausibly produce the reported results on its own: (i) Positives and negatives in Dataset 2 come from different sources, so the classifier may be separating provenance rather than calls (Comment 1); this requires same-source negatives. (ii) The cross-validation may leak at the recording level (Comment 2); this requires grouped, leave-recording-out partitioning. (iii) The baseline and the classifiers are compared at different class prevalences, so the reported gain cannot be attributed to the classifier (Comment 3); this requires matched prevalence.

I recommend major revision. Several comments below are requests for clarification rather than assertions, because the manuscript does not fully specify some relevant details, for example how the cross-validation folds were partitioned. If the author has already controlled for these issues, the manuscript should say so explicitly.

Should the evaluation be revised so that it isolates the secondary classifier's true contribution, and the claims calibrated to the resulting evidence, this would be a useful addition to the bioacoustic monitoring literature, and, in my view, a more valuable one than the current framing achieves, since a well-characterised realistic result serves practitioners better than an unqualified perfect score.

Major comments

1. Positives and negatives in Dataset 2 come from different sources (Section 2.1.2).

In Dataset 2, the target samples are drawn from Holderried (2024), while the non-target samples are the author's own Xeno-canto-derived negatives, reused from Dataset 1. Positives and negatives therefore originate from entirely different recording campaigns, with likely systematic differences in recording equipment, sites, background soundscape, and audio encoding.

This means a classifier achieving $AP \approx 1.0$ on this split may be separating recording provenance, Holderried recordings versus Xeno-canto negatives, rather than woodcock call versus no call. Because Dataset 2 supplies the manuscript's strongest and most prominently reported result, I regard this as a concern that must be addressed before the result can be fully interpreted.

I would ask the author to either (i) evaluate against negatives drawn from the same source and, ideally, the same recordings as the Holderried positives, or (ii) state the confound explicitly and temper all conclusions drawn from Dataset 2.

2. Possible recording-level leakage in the cross-validation (Section 2.2).

In Dataset 1, the 501 positive and 2505 negative segments are all extracted from the same 97 recordings. The manuscript describes a 5-fold cross-validation but does not state whether folds were partitioned **at the recording level or at the segment level**.

If the split is at the segment level, segments originating from the same recording, and therefore sharing an individual, site, microphone, and noise floor, will appear in both training and test partitions. The classifier can then achieve high scores by recognising recording-specific signatures rather than the species' call. This is a documented pitfall in bioacoustic classification, and it would be a parsimonious explanation for an AP of 1.000 accompanied by almost no variance across folds.

I would state explicitly how the folds were partitioned. I think that, if the partitioning was segment-level, the analysis should be repeated using grouped cross-validation or it should be mentioned as a possible limitation.

3. The baseline and the secondary classifiers are not evaluated under comparable conditions.

The BirdNET baseline AP (84.3% and 89.3%) is computed on imbalanced data, 501 positives against 2505 negatives, approximately 1:5. The secondary-classifier AP (99.8-100%) is computed on deliberately balanced folds of 501 against 501.

Average precision is prevalence-dependent: precision falls as the negative-to-positive ratio rises, because false positives accumulate with the number of negatives. An unknown share of the apparent improvement from ~84% to ~100% can therefore be attributable to the change in class balance rather than to the secondary classifier. As

the experiment is currently designed, the contribution of the classifier cannot be separated from the contribution of the evaluation protocol.

To isolate the classifier's actual effect, I think the BirdNET thresholding baseline and the secondary classifiers should be evaluated under the same cross-validation protocol and at the same class prevalence. I would encourage the author to report the baseline on the balanced folds as well, since that comparison is straightforward to produce and would strengthen the paper considerably if the gain survives it.

4. The evaluation does not reflect the deployment regime that motivates the work.

The manuscript itself cites Engler et al. reporting a BirdNET precision of 22.0%, 1475 true positives from 6803 hits, in a genuine field deployment. That figure illustrates the real bottleneck in passive acoustic monitoring: very low target prevalence across large volumes of continuous audio, generating large numbers of false positives.

A classifier trained and tested on a balanced, curated set of clean positive and negative segments does not speak to that regime. Yet the manuscript's conclusions, "reliable automated bioacoustic monitoring tools", "near-perfect identification", imply that it does. The paper reports a field precision of 22.0% and a benchmark precision of more or less 100% within a few pages of each other, without reconciling them. I would ask the author either to add an evaluation under deployment-realistic conditions (low prevalence, unseen sites, continuous recordings) or to soften these claims.

5. The "unseen" / "independent evaluation" framing needs substantiation (Section 2.1.1).

The claim that Dataset 1 constitutes an "unseen evaluation dataset" for BirdNET rests on the recordings dating from 2023-2024. I think two problems follow.

First, the manuscript never states which BirdNET version was used, nor its training cutoff, so the claim of unseen data is unverifiable as written. Second, recordings drawn from Xeno-canto, the repository BirdNET was substantially trained on, as the author notes in Section 2.1, are by construction domain-similar rather than out-of-distribution, even where the specific files were withheld. Withholding files from a distribution is not the same as testing outside that distribution.

Please state the BirdNET version and, where possible, its training cutoff, and reframe Dataset 1 as "domain-similar held-out data" rather than as an "unseen" or "independent evaluation set", unless out-of-distribution status can be positively demonstrated.

6. "Linear probing" does not describe two of the three methods.

Linear probing conventionally denotes a *linear* classifier fitted on frozen embeddings. Of the three classifiers used here, only the SVM is described as linear-kernel; Random Forest and XGBoost are non-linear. The title's claim that "linear probing enables..." therefore does not describe the method actually reported, and two-thirds of the results fall outside this framing. I would suggest either adopting more general terminology such as "posterior classifiers" or restricting the linear-probing framing to the SVM results.

Also, Section 2.3 states that “spectrogram images are used as input data”, which contradicts the Abstract, Fig. 2, and the rest of the manuscript, where BirdNET *embeddings* are the classifier inputs. Can you resolve this inconsistency?

Minor comments

7. Report variance and uncertainty. AP values of 1.000 are reported without confidence intervals or any measure of dispersion, despite 5 folds × 5 splits being available. Please report mean ± SD (or CIs) for the metrics.

8. BirdNET version and inference settings. Overlap, segment length, sensitivity, and confidence handling are not specified anywhere in the manuscript. These are necessary for reproducibility and for interpreting Figs. 3-5.

9. Repository links are inconsistent. Two different GitHub URLs are given: .../WoodcockCNN in the Acknowledgements and .../Woodcock-CNN under Code availability. Please provide a single working link.

10. Fig. 1 cannot show the calls as described. The spectrogram’s frequency axis extends only to approximately 10 kHz, whereas the text (Section 1) states that the squeak reaches a maximum frequency of up to 20 kHz. The figure therefore cannot display the full bandwidth of the call it is intended to illustrate. It could be nice to extend the axis. The figure is also a low-resolution screenshot including the Audacity interface bar; a cleaner, higher-resolution rendering would be preferable for publication.

11. Fig. 2 does not depict the pipeline actually used. The schematic labels the feature extractor generically as “Foundational Model (e.g., Bird Audio Encoder)” rather than BirdNET, and the illustrative spectrogram does not correspond to a woodcock recording. Since the figure is presented as the methodology of this study, it should depict the specific pipeline used. The presence/absence legend is also duplicated, it appears both within panel 3 and in a separate box below, and could be given once. Finally, the species name in the figure should be italicised (*Scolopax rusticola*), consistent with the text. Also panel 3 is headed “Linear Probe (Binary Classification)” but contains a box labelled only “ML Classifier”. This reproduces the terminological problem raised in Comment 6: two of the three classifiers actually reported (Random Forest, XGBoost) are not linear probes, so the figure asserts something the Results contradict. Since the figure is presented as the methodology of this study, the label should match the methods actually used.

Typos

13. p. 7: “a maximum value of 0.833 is obtained for **Dataset.**” the dataset number appears to be missing (“Dataset 2”).

14. Section 2.1.1: correct “guaranties” in “guarantees”.

15. Italicisation of *Scolopax rusticola* is inconsistent throughout the manuscript and figures.

16. The reference to Engler et al. appears without a year in several places in the text (“Engler et al.” in Section 1 and Section 3.1) while being cited as “Engler et al. (2025)” elsewhere; the reference list entry also lacks a year.

Reviewer 3

This manuscript addresses a problem of real practical importance to the passive acoustic monitoring community: how to improve species-specific detection performance when a general-purpose classifier such as BirdNET is applied to a single target species. The author extracts BirdNET embeddings for 3-second audio segments and trains three second-stage binary classifiers (SVM, Random Forest, XGBoost) to discriminate Eurasian Woodcock *Scolopax rusticola* calls from non-target sounds.

The underlying idea is well conceived and timely. Reusing frozen embeddings from a large bioacoustic model and fitting a lightweight species-specific classifier on top is exactly the kind of pragmatic approach that makes automated monitoring tractable for ecologists working on a single focal species, and the author has framed the motivation clearly with a thorough and well-chosen literature base. The choice of Eurasian Woodcock is a good one: the species is genuinely difficult to survey, the roding display gives a well-defined acoustic target, and there is an active applied need for reliable detection.

My main concern is that the headline result: near-perfect average precision is very likely to be optimistic as a consequence of how the evaluation is constructed, rather than a reflection of what the method would achieve in deployment. I want to stress that this looks entirely fixable, and that a revised, more conservative set of numbers would in my view make for a stronger and more useful paper. Detection performance in the 0.90s would be a genuinely valuable result for the community.

1. Probable information leakage between training and test partitions

Source confounding in Dataset 2. The positive samples in Dataset 2 originate from Holderried (2024), while the negative samples are drawn from the author’s own Xeno-canto-derived material. A classifier presented with this combination can separate the two classes on the basis of recording provenance, recording equipment, sample rate and resampling history, background noise floor, reverberation characteristics, habitat, or geography, without learning anything about woodcock vocalisations at all. Because BirdNET embeddings encode a good deal of general acoustic scene information, this is not a remote possibility. As currently constructed, I do not think the Dataset 2 results can be interpreted as evidence about species detection.

Possible remedy: rebuild Dataset 2 so that positives and negatives are drawn from the

same recording pool. If the negatives from Holderried cannot be released, it may be possible to negotiate access for evaluation purposes, or alternatively to source negatives that match the positives in recorder type, location and season. If neither is feasible, a direct empirical check would help: train a classifier to predict source dataset rather than species, and report its accuracy. A high value would confirm the confound; a low one would substantially strengthen the author's case.

2. The baseline and the second-stage classifiers are not evaluated on comparable terms

The BirdNET baseline (84.3% / 89.3% AP) is measured over the full annotated material, whereas the second-stage classifiers are evaluated on balanced, under-sampled partitions in which negatives are drawn from the same recordings as the positives. The improvement narrative therefore compares two rather different tasks, and the magnitude of the gain is likely to be inflated by that difference rather than by the method alone.

Suggested remedy: define a single held-out test set, recording-independent, and with a positive-to-negative ratio representative of real monitoring data and score both BirdNET and each second-stage classifier on it. Under-sampling is a reasonable device for training under class imbalance, but the test set should reflect deployment conditions. This single change would make the central claim of the paper much more defensible.

3. Negative sampling and the scope of the generalisation claim

Discriminating woodcock calls from other segments of the same recordings is a considerably easier problem than discriminating them from the full range of sounds encountered in continuous forest monitoring: sympatric species, wind and rain, anthropogenic noise, and as Engler et al. (2025) show for wind farm settings, acoustic conditions in which BirdNET's precision falls dramatically, to 22%. The manuscript would be strengthened by acknowledging this explicitly and, ideally, by evaluating on at least a modest sample of continuous soundscape audio. Even a small-scale demonstration on unsegmented field recordings would considerably increase the practical credibility of the approach for the readers most likely to want to use it.

4. Terminological framing of "linear probing"

The term is conventionally reserved for a linear classifier fitted to frozen embeddings. This describes the linear-kernel SVM well, but Random Forest and XGBoost are non-linear models and are not linear probes. Since the SVM is in fact the best performer, an option would be to foreground it as the linear probe and present the tree ensembles as non-linear comparators. This would preserve the framing, tighten the narrative, and make the interesting observation that the linear model wins into a substantive point about the geometry of the BirdNET embedding space rather than an incidental result.

5. Reconciling Figure 2 and Section 2.3 with the implemented method

Section 2.3 and the schematic in Figure 2 describe spectrogram images as the input

representation, whereas the method as implemented passes audio to BirdNET and uses the resulting embeddings. A reader might be uncertain about what the classifier actually receives. I would suggest redrawing the figure to show the true pipeline (audio → BirdNET encoder → embedding vector → binary classifier) and adjusting the accompanying text accordingly. I would also gently suggest that a figure carrying the paper's core methodological explanation may merit being drawn by hand rather than generated by an AI tool, given how much explanatory weight it bears.

6. Scope of the title

The title refers to near-perfect identification of “bird species” in general, whereas the study evaluates single-species binary detection for one taxon under the evaluation conditions discussed above. Narrowing the title to reflect the actual scope, naming the species, and framing the contribution as detection rather than identification would be more appropriate.

Minor comments

Reproducibility details. Please state the BirdNET version used, which layer the embeddings were taken from and their dimensionality, and whether the 3-second annotation windows align with BirdNET's native analysis window. Details of any resampling, normalisation or overlap between adjacent windows would also help. Similarly, the classifier hyperparameters (SVM kernel and C, forest size and depth, XGBoost settings) should be reported, even if defaults were used.

Class definition. Grunts and squeaks differ substantially in their acoustic properties. Were both pooled into a single positive class? If so, it would be interesting to know whether performance differs between call types, this is likely to matter for field application, since the two carry different detection ranges.

Copy-editing. A few items to pick up at revision:

Section 2.1.1: “guaranties” → “guarantees”.

Section 3.1: “a maximum value of 0.833 is obtained for Dataset.” — the dataset number is missing.

Reference list: the journal abbreviated as “arla” for the Pérez-Granados entries should be Ardeola.

Capitalisation of *Scolopax rusticola* and of “Eurasian Woodcock” varies through the text; please standardise.

Two GitHub URLs are given in the Acknowledgements and the Code availability statement

respectively, with slightly different formatting — please confirm they resolve to the same repository.

“an Eurasian Woodcock” (Fig. 1 caption) → “a Eurasian Woodcock”.